A Later Small-Radii Literature Answer for the Discrete Euclidean Ball Maximal Problem

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Status. This is a lightweight literature-implied status note:

`literature_implied_answer (partial small-radii subrange).`

It records later progress on the discrete Euclidean ball maximal-function problem highlighted in Bourgain–Mirek–Stein–Wrobel, *On the Hardy–Littlewood maximal functions in high dimensions: Continuous and discrete perspective*, arXiv:1812.00153.

Source problem. In the introduction and in Section 3, "Discrete perspective", the source studies whether the discrete Hardy–Littlewood maximal constant

$$\mathcal{C}_p(d, G)$$

for averages over $G_t \cap \mathbb{Z}^d$ can be bounded independently of d. For $G = B^2$, Theorem 1.2 of the source proves a dimension-free large-scale estimate for

$$\sup_{t \geq Cd} |\mathcal{M}_t^{B^2} f|.$$

The source then says that the matter is reduced to understanding $\sup_{1 \leq N \leq Cd} |\mathcal{M}_N^{B^2} f|$. It records the dyadic ball estimate for $p \in [2, \infty]$, but states that the available multiplier bounds do not give the full maximal function; in particular, obtaining the third discrete multiplier estimate in equation (49) would be a route toward dimension-free estimates for $\mathcal{C}_p(d, B^2)$.

Supporting literature. Niksinski–Wrobel, *Dimension-free estimates for discrete maximal functions and lattice points in high-dimensional spheres and balls with small radii*, arXiv:2503.16952, gives a later full-maximal small-radii result for the Euclidean ball. Its Theorem 1.1 states that for every $\varepsilon > 0$ there is C_ε such that, uniformly in d and $p \in [2, \infty]$,

$$\left\| \sup_{0 \leq t \leq d^{1/2-\varepsilon}} |\mathcal{M}_t f| \right\|_{\ell^p(\mathbb{Z}^d)} \leq C_\varepsilon \|f\|_{\ell^p(\mathbb{Z}^d)}.$$

This is a direct positive answer on a non-dyadic small-radii subrange of the source problem. Remark 1 of the supporting paper explicitly combines this small-radii theorem with the large-scale theorem from the Bourgain–Mirek– Stein–Wrobel survey to say that the remaining range relevant to Stein's discrete question is

$$t \in (d^{1/2-\varepsilon}, Cd).$$

Scope limitations. This packet is not a new proof and not a full settlement. It records a later partial literature answer. The middle range

$$d^{1/2-\varepsilon} < t < Cd$$

remains outside this identification, as does removal of the ε loss. The supporting theorem is also in the range $p \in [2, \infty]$, so it does not settle the broader $p \in (3/2, \infty]$ or $p \in (1, \infty]$ possibilities discussed in the source.

Search evidence. Cheap local indexes had no exact prior row for arXiv:1812.00153. A web/arXiv search for the exact paper and the discrete Euclidean-ball full maximal problem identified arXiv:2503.16952 as the later small-radii result. The parsed local source and downloaded supporting PDF were checked for the source's Section 3 reduction and for Theorem 1.1 plus Remark 1 in the supporting paper.

References.

- J. Bourgain, M. Mirek, E. M. Stein, and B. Wrobel, *On the Hardy–Littlewood maximal functions in high dimensions: Continuous and discrete perspective*, arXiv:1812.00153, v2, 2019.
- J. Niksinski and B. Wrobel, *Dimension-free estimates for discrete maximal functions and lattice points in high-dimensional spheres and balls with small radii*, arXiv:2503.16952, v2, 2026.